UNIVERSITY OF BIRMINGHAM

School of Computer Science

Week 5 Exercise Questions

Artificial Intelligence 2

2024

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Exercise questions Week 5

Question 1

Carnegie Mellon University conducted a comprehensive study on students' happiness by asking the following yes-or-no questions:

- Do you party frequently [Party: Yes/No]?
- Are you wicked smart [Smart: Yes/No]?
- Are you creative [Creative: Yes/No]?
- Did you do well on all your continuous assessment? [HW: Yes/No]
- Do you use a Mac? [Mac: Yes/No]
- Did your final project succeed? [Project: Yes/No]
- Did you succeed in your final year project? [Success: Yes/No]
- Are you currently Happy? [Happy: Yes/No]

You can obtain the survey results (students.csv) from here. Each row in students.csv corresponds to the responses of a separate student. The columns in students.csv correspond to each question (random variable) in the order Party, Smart, Creative, HW, Mac, Project, Success, and Happy. The entries are either zero, corresponding to No response, or one, corresponding to a Yes response. After consulting a behavioural psychologist we obtained the following complete set of conditional relationships:

- HW depends only on Party and Smart
- Mac depends only on Smart and Creative
- Project depends only on Smart and Creative
- Success depends only on HW and Project
- Happy depends only on Mac, and Success

Complete the following questions:

(a) Draw the Bayesian network that represents the relationship as described above.

(b) Write joint distribution as a product of conditional probabilities.

(c) What is the number of independent parameters needed for each conditional probability table?

(d) What is the total number of independent parameters?

0 marks for question not valid - all questions must have the same number of marks

Question 2 (2022 AI2 Exam question)

As a machine learning expert for an AI cyber security company, your task is to design an automated network intrusion detection system. You have collected a large number of records of network activities. Each record includes the log information about network activity, such as protocol types, duration, number of failed logins, which are random variables, denoted as $\boldsymbol{X} = [X_1, X_2, \ldots, X_n]^\intercal$. Each record also includes a binary random variable Y called label that was labelled by cyber security experts as intrusions ($Y = 1$) or normal connections ($Y = 0$).

(a) Omitted

(b) After applying your feature selection algorithm, assume you selected four random variables as features, denoted as F_1, F_2, F_3, F_4. Based on these features, you now work with a cyber security expert to construct a Bayesian network to harness the domain knowledge of cyber security. The expert first divides intrusions into three cyber attacks, A_1, A_2, A_3, which are marginally independent from each other. The expert suggests the presence of the four features are used to find the most probable type of cyber attacks. The four features are conditionally dependent on the three types cyber attacks as follows: F_1 depends only on A_1, F_2 depends on A_1 and A_2. F_3 depends on A_1 and A_3, whereas F_4 depends only on A_3. We assume all these random variables are binary, i.e., they are either 1 (true) or 0 (false).

   (i) Draw the Bayesian network according to the expert?s description. **[2 marks]**

   (ii) Write down the joint probability distribution represented by this Bayesian network. **[3 marks]**

   (iii) How many parameters are required to describe this joint probability distribution? Show your working. **[5 marks]**

   (iv) Suppose in a record we observe F_2 is true, what does observing F_4 is true tell us? If we observe F_3 is true instead of F_2, what does observing F_4 is true tell us? **[5 marks]**